Costain Nachuma

Software Engineer | Backend Systems & Developer Tooling
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SUMMARY

Software engineer and Ph.D. student. Shipped **Comprendia**, a public Eclipse plugin that grounds LLM code explanations in a live dependency graph, and the pipelines behind two MSR papers: 33,596 agent-authored pull requests and 14.4 million Maven releases. Nine years of industry engineering, including a country-wide provisioning system that cut configuration time 90%.

TECHNICAL SKILLS

Languages	Python, Java, C#, JavaScript, SQL
Systems	Linux, Git, Docker, GitHub Actions, Eclipse JDT, JGit
Data	DuckDB, Neo4j, PostgreSQL, MySQL, Oracle, Parquet, Pandas
Applied	LLM APIs (Claude, OpenAI, Ollama), CVE/OSV analysis, SBOM evaluation

EXPERIENCE

Graduate Research & Teaching Assistant — 2023 – Present

Idaho State University, Pocatello, ID

- Shipped **Comprendia**, a five-layer Eclipse plugin (Cytoscape.js UI, JDT AST engine, OSV.dev/JGit vulnerability overlays, and an `AiGateway` across Claude, OpenAI, and Ollama). **Graph-Aware Callee Pruning** keeps prompts on the live call graph with an $O(V+E)$ filter, so explanations stay reproducible across model families. Benchmarked on 6 fixtures × 4 prompt modes × 3 LLM families. Tool demo at ICSME 2026. Artifact | Demo.
- Built a DuckDB pipeline over **33,596** agent-authored pull requests (2,807 repositories, 5 coding agents). Reviewer engagement was the strongest predictor of a merge; overall merge rate **71.5%**. Published at MSR 2026.
- Built a Neo4j pipeline over **14,459,139** Maven releases (658,078 artifacts), a 21-hour job on a 64GB, 24-core machine. **62.89%** of the latest releases carry a transitive vulnerability, against 31.39% direct. Published at MSR 2025. Mentored 20+ students.

Instructor of Record, CS 3309 — Fall 2025

Idaho State University, Pocatello, ID

- Sole instructor for 45 undergraduates: curriculum, lectures, and grading. Teaching effectiveness 3.50/4.0, with 90% agree or strongly agree.

Senior Support Specialist, Software Engineer — 2020 – 2023

National Assembly of Zambia, Lusaka

- Modernized parliamentary web platforms and designed the test protocol for sessions with the VP, ministers, and MPs. Pre-meeting error discovery rose to about 95%.

Software Developer, Technical Support Specialist — 2012 – 2018

National Assembly of Zambia, Lusaka

- Wrote an automated provisioning script for a country-wide network: configuration time down 90%, results consistent, issue resolution down 30%. Supported 500+ staff.

SELECTED PROJECT

- **Chromium bug ETL** (Python, BeautifulSoup). Five scripts against the live Chromium tracker, past a 1,000-record API cap: 5,575 security and 4,605 performance bug reports.

EDUCATION

Ph.D., Computer Science, GPA 4.0 — 2023 – Spring 2028

Idaho State University | AI-assisted code comprehension and review

M.Sc., Computer Science — 2018 – 2020

University of New Orleans

B.Sc., Computer Science — 2007 – 2011

University of Zambia, Lusaka